

PROFESSIONAL PROFILE

I combine deep pre-AI systems engineering (5+ years in C/C++, Linux kernel internals, DPDK/VPP packet processing, and hardware) with modern AI tools (OpenAI Codex Desktop, DeepMind AGY CLI). In my projects, AI is never a fragile prompt gimmick; it is an intelligent cognitive brain orchestrated by a robust, deterministic software layer and production bare-metal Linux infrastructure.

CORE COMPETENCIES & TECHNICAL STACK

Systems & Low-Level: C, C++20, Linux Kernel, Device Drivers, MIPS, STM32, DPDK, VPP, DPI, conntrack, POSIX, CMake, Make	Backend & Databases: Python (FastAPI, asyncpg, Pydantic), Node.js (Fastify, Express), PostgreSQL (18+ migrations), Redis, SQLite
Frontend & Mini Apps: React 19, TypeScript, Telegram Mini Apps API, Tailwind CSS, Vite, Framer Motion, SEO, Core Web Vitals	Infrastructure & AI: Bare-Metal Ubuntu, Docker, Docker Compose, Nginx, Caddy, Cloudflare Tunnels, systemd, Codex Desktop, AGY CLI

SELECTED FEATURED PROJECTS

Procurement AI Agent & Tender Helper — Autonomous B2B procurement & tender orchestrator (Telegram, WhatsApp, AGY CLI)	AGY CLI (Gemini), Telegram Bot, WhatsApp (WAHA), Python
Telegram Ingress + durable PostgreSQL outbox/leases (SKIP LOCKED) + sandboxed AGY CLI (Google Antigravity) with strict JSON Schema. WhatsApp (WAHA) boundary guarded by human-in-the-loop approval gates (immutable payload hash verification). AssemblyAI bilingual RU/KK voice transcription, fail-closed Decimal-only calculations.	
Shermos Bot & Mini App — Telegram service & Mini App for manufacturing company in Bishkek	FastAPI, PostgreSQL, Redis, React 19
Webhook + Python worker architecture with FastAPI, asyncpg, Pydantic. Distributed Redis queues with per-user mutex locks. 18 PostgreSQL migrations. 3D rendering pipeline. Telegram Mini App in React 19, TypeScript, and Vite.	
Musikschule Bielefeld & AI Telegram CMS — European web platform with autonomous AI-driven Telegram CMS	React 19, Telegram AI CMS, AGY CLI, AssemblyAI
React 19 + Vite + Tailwind frontend. Backend features an AI Telegram bot powered by Google Antigravity (AGY CLI 1.2.7 with Gemini 3.8 Flash) and AssemblyAI universal-3-5-pro. The agent parses voice notes, edits repository code, converts assets to WebP, generates a Git diff for admin approval, and executes zero-downtime Docker redeployments with HTTP origin verification upon /confirm. Enforces self-contact phone allowlist authorization.	
Floorball KZ & Content Agent Bot — Official Kazakhstan Floorball Federation portal & Telegram Mini App CMS	React, Telegram Mini App, Agy CLI, PostgreSQL
React + TypeScript web portal. Content ingestion backend: Python Telegram bot + background worker, PostgreSQL durable queue/outbox, AssemblyAI bilingual RU/KZ speech-to-text, Agy CLI. Integrated Telegram Mini App with initData signature verification, multi-tier RBAC for city coaches, news galleries up to 10 photos, automated multi-device screenshot preview, and approval-gated atomic Git push.	
Emily Brautmoden Boutique — Premium bridal boutique redesign & commercial showcase in Germany	React 19, TypeScript, Vite, i18n
React 19, TypeScript, Vite, modular CSS with custom token-driven design system, Framer Motion, react-i18next (DE/EN/RU), react-router-dom v7, Core Web Vitals optimization (LCP, CLS), deployed on Vercel/Netlify.	

PROFESSIONAL WORK EXPERIENCE

Software Engineer / Independent Contractor | Freelance & Own Projects

Mar 2025 – Present · Remote (Clients from Germany, Kazakhstan, Kyrgyzstan)

- Infrastructure provisioning: AWS EC2, bare-metal Ubuntu servers, Docker / Docker Compose, Nginx, Caddy, systemd units.
- Networking & Security: automated Let's Encrypt (ACME) certificates, Cloudflare tunnels, HTTP/2, HTTP/3, and API edge protection.
- DevOps workflows: idempotent Bash deployment automation, automated PostgreSQL migrations, monitoring, and production profiles.
- Built production Telegram Mini Apps with FastAPI, asyncpg, Redis distributed locks, and React 19.

Programming Languages Instructor (C / C++) | Luxkod

Oct 2023 – Aug 2024 · Kazakhstan

- Authored and delivered hands-on C and C++ curricula: pointers, memory management, data structures, algorithms.
- Guided students through compiler error diagnosis, memory leak profiling (Valgrind, GDB), and modular project architecture.

Software Engineer / Automation Manager | TOO Zavod Armatura Taraz

Jan 2023 – Oct 2023 · Taraz, Kazakhstan

- Engineered a dedicated C++ application for automated extraction and generation of production reports from Excel datasets.
- Created VBA automation pipelines for inventory reconciliation, logistics manifests, and delivery contracts.
- Streamlined end-to-end document workflows for cargo logistics and warehouse distribution.

C Developer / Systems Engineer | Ideco

Nov 2022 – Dec 2022 · Ekaterinburg

- Developed C systems software leveraging DPDK and VPP (Vector Packet Processing) network stacks.
- Engineered control-plane API extensions and automated unit tests for DPI (Deep Packet Inspection) modules.
- Investigated and resolved bugs in traffic-filtering engines: IPS, DPI, and Linux kernel conntrack.
- Maintained CMake build systems and collaborated in Jira and Git.

Systems & Embedded Software Engineer | In-Soft LLC (Eltex)

Aug 2021 – Oct 2022 · Novosibirsk

- Patched Linux kernel and developed hardware device drivers to expand wireless device capabilities.
- Authored userspace utilities and Bash scripts for dynamic access point configuration.
- Implemented networking services: DHCP, RADIUS, TFTP, SSH, Docker environments, and config parsing via json-c.

Laboratory Assistant, 3rd Category | Budker Institute of Nuclear Physics (BINP SB RAS)

Oct 2020 – Aug 2021 · Novosibirsk

- Programmed STM32 microcontrollers in C for real-time digital noise filtering from electret microphones.
- Implemented digital signal processing algorithms interfacing with sensitive analog sensors.

EDUCATION

Novosibirsk State University (NSU)

2018–2021 | Новосибирск

Physical & Technical Informatics (2018–2021) · Strong mathematical & physics foundation